Ex.no: 11
Date: 30-9-24

11. A) SIMULATE STATIC ROUTING PROTOCOL CONFIGURATION

Aim:

To configure and simulate static routing with main and floating backup routes.

Procedure:

step 1: open cisco packet tracer and create the required network topology using 3 routers.

step 2: connect the routers and PCs using suitable cables.

step 3: configure the IP addresses and subnet masks on the router interfaces

step 4: Activate the interfaces using the no shutdown command.

step 5: configure static routes using the ip route.

step 6: Assign a lower AD to the main route and higher AD to the backup route.

step 7: Verify the routing table using the show ip route static.

step 8: Test connectivity using the ping and tracert.

step 9: Disconnect the main link and verify whether the backup route is selected.

Result:

Thus, static routing with main and backup routes was successfully simulated and verified.

11. b) SIMULATE RIP USING CISCO PACKET TRACER

Aim:

To simulate the dynamic routing information protocol (RIP) using cisco and verify dynamic routing between different networks.

RIP:

Routing Information protocol is a dynamic distance vector routing protocol that uses hop count as a routing metric to determine the best path b/w source and destination networks.

Procedure:

step 1: open cisco packet tracer and create a topology with 3 routers, 2 PCs.

step 2: connect the devices using suitable Ethernet and serial cables.

step 3: configure IP addresses on PC0 and PC1

step 4: Activate all router interfaces using the no shutdown.

step 5: Enable RIP on each router interfaces using the router rip command.

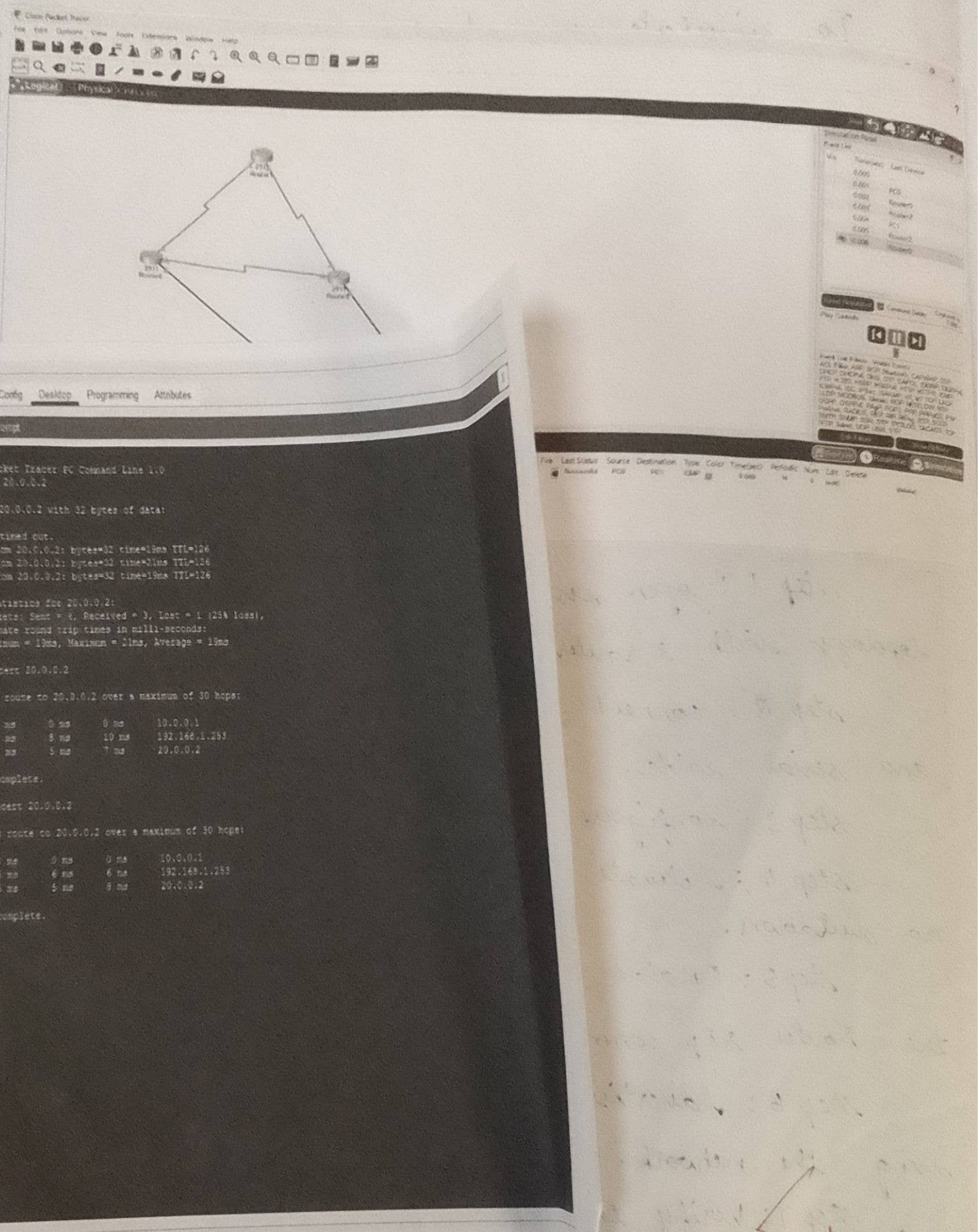
step 6: Advertise the directly connected networks using the network.

step 7: Verify the routing information using the show ip route.

step 8: The best connectivity between PC0 and PC1 using the ping.

step 9: Use tracert to observe the route taken by packets.

Step 10: Disconnect the direct link between Router 0 and router 2 and verify the alternate route through router 1.



Result:

This, dynamic routing using RIP was successfully simulated and verified